

CSC241 - OBJECT ORIENTED PROGRAMMING (JAVA)

Pre-Lab Environment Setup Guide

For CS and AI Sections

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ARRIVE READY - NOT READY TO DOWNLOAD

Bring your own laptop and charger. Before the 3-hour lab, Java must compile and run from Command Prompt, Git must work, and you must be able to sign in to a verified GitHub account. Use lab time for programming and Git/GitHub practice, not basic installation.

1 Mandatory preparation before the lab

Requirement	Ready when...
Laptop + charger	You can use the laptop in class and have permission/admin access to install software.
Working Java JDK	java and javac both work in Command Prompt. Fall 2026 class standard: JDK 25 LTS.
Git installed	git --version prints a version.
GitHub account	Email verified and sign-in tested before the lab.
Local course workspace	Your registration-number folder exists under C:\CSC241.

JAVA VERSION POLICY - UPDATED 10 SEPTEMBER 2026

Use the latest supported LTS JDK adopted for the course, not an old Java release. For Fall 2026, use JDK 25 LTS. JDK 26 is newer, but it is a feature release and is not required for the class. Install a full JDK, not only a JRE, because the javac compiler is required.

OFFLINE PREPARATION TIP

Download/save this PDF on your laptop or phone. Initial software download and GitHub account creation need internet access, but Java compile/run checks can be repeated offline afterward.

2 Install Java from Command Prompt (Windows)

Open Command Prompt (as Administrator if required). First check Windows Package Manager:

CHECK WINGET

```
winget --version
```

Install Eclipse Temurin JDK 25 LTS:

INSTALL JAVA JDK 25 LTS

```
winget install EclipseAdoptium.Temurin.25.JDK
```

Close Command Prompt and open a new one. Verify both runtime and compiler:

VERIFY JAVA

```
java --version
javac --version
where java
where javac
```

PASS CONDITION

Both java --version and javac --version must succeed. Patch numbers may change as security updates are released; the major JDK family should match the course-approved version.

3 Create your permanent local CSC241 workspace

Keep your course work in one predictable location on drive C:. This folder will later be suitable for Git/GitHub synchronization.

Windows naming rule: your registration number contains /, which cannot be used in a Windows folder name. Replace every slash with a hyphen.

EXAMPLE

Registration: CIIT/SP26-XXX-001/LHR

Folder: C:\CSC241\CIIT-SP26-XXX-001-LHR

Run these commands, replacing the example registration with your own:

CREATE THE COURSE FOLDER

```
cd /d C:\
mkdir CSC241
cd C:\CSC241
mkdir CIIT-SP26-XXX-001-LHR
cd CIIT-SP26-XXX-001-LHR
mkdir Lab01
cd Lab01
cd
```

Optional check:

VIEW THE FOLDER TREE

```
tree C:\CSC241 /F
```

KEEP THIS LOCATION CONSISTENT

Do not scatter copies of the same lab work across Desktop, Downloads and random folders. Keep the registration-number folder as your permanent CSC241 workspace. The GitHub remote connection will be taught/practiced in the lab.

4 Compile and run a Java program without an IDE

Inside your Lab01 folder, create a plain-text file named exactly HelloWorld.java:

HELLOWORLD.JAVA

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("CSC241 Java setup is working!");  
    }  
}
```

In Command Prompt, make sure you are in the same folder, then:

COMPILE AND RUN

```
dir  
javac HelloWorld.java  
dir  
java HelloWorld
```

EXPECTED RESULT

The compiler creates HelloWorld.class. Running java HelloWorld prints CSC241 Java setup is working!. This is stronger evidence of a correct setup than only checking java --version.

5 Install Git from Command Prompt (Windows)

Install Git for Windows using winget:

INSTALL GIT

```
winget install --id Git.Git -e --source winget
```

Close/reopen Command Prompt and verify:

VERIFY GIT

```
git --version  
where git
```

Configure the identity Git will attach to your commits. Use your own name and an email associated with your GitHub account:

ONE-TIME GIT IDENTITY SETUP

```
git config --global user.name "Your Full Name"  
git config --global user.email "your-email@example.com"  
git config --global --list
```

DO NOT COPY THE SAMPLE EMAIL

Enter your own email address. Do not connect/push the local folder to a GitHub remote yet unless your instructor asks you to; that workflow will be practiced in the lab.

6 Create and verify your GitHub account

1. Create/sign in: <https://github.com/signup> - if you already have an account, keep using it.
2. Use a professional username: choose one appropriate for coursework and future professional use.
3. Verify your email: complete the verification email before the lab.
4. Test sign-in: sign out and sign in once before coming to class. Do not arrive needing password recovery.
5. Secure the account: use a unique password and enable two-factor authentication when possible. Never share passwords or verification codes.

7 Manual installation fallback

If winget is unavailable or blocked, use the official installers:

Java JDK	https://adoptium.net/installation/ - select the current course-approved LTS JDK (Fall 2026: JDK 25 LTS).
Git for Windows	https://git-scm.com/install/windows - choose the maintained build appropriate for your Windows system.
GitHub account	https://github.com/signup

8 Quick troubleshooting

Problem	What to do
winget not recognized	Use the official manual installers above.
java / javac not recognized	Open a new Command Prompt. Run where java / where javac. Reinstall the full JDK or correct PATH.
Wrong Java version appears	Run where java and where javac; another Java installation may appear earlier on PATH.
Could not find/load main class	Be in the folder containing HelloWorld.class and run java HelloWorld without .class.
git not recognized	Open a new Command Prompt, run where git, and reinstall if needed.
GitHub sign-in problem	Finish email verification/password recovery before the lab.

9 Final readiness check - complete before leaving home

[]	I have my laptop and charger.
[]	java --version works.
[]	javac --version works.
[]	I compiled HelloWorld.java and ran java HelloWorld successfully.
[]	git --version works.
[]	My Git name and email are configured correctly.
[]	My GitHub account is verified and I can sign in.
[]	My permanent local folder exists under C:\CSC241\.
[]	Lab01 exists inside my registration-number folder.
[]	I saved this PDF locally for offline reference.

READY FOR LAB = FOUR THINGS WORK

1) Java compiles and runs from Command Prompt. 2) Git works. 3) GitHub login works. 4) Your permanent registration-number workspace is ready on drive C:.

10 Command cheat sheet

MOST-USED SETUP COMMANDS

```
winget install EclipseAdoptium.Temurin.25.JDK
java --version
javac --version
winget install --id Git.Git -e --source winget
git --version
cd /d C:\CSC241
tree C:\CSC241 /F
```

SOURCE / VERSION NOTE

Revised 10 September 2026. At that time, JDK 25 was the current Java LTS family and JDK 26 was a current feature release. Eclipse Adoptium documented the Windows winget command for Temurin 25; Git for Windows documented its winget installation command. Follow later instructor guidance if the class standard changes.

Official references: <https://adoptium.net/installation/> | <https://git-scm.com/install/windows> | <https://github.com/signup>